

Jeremy Wayland, PhD

Founder | Machine Learning Researcher

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Machine learning researcher and founder specializing in geometric deep learning and clinical AI applications. PhD in mathematics and machine learning with the AIDOS Lab at TU Munich. Now Co-Founder of Krv Labs, building patient-discovery infrastructure for clinical trials. Dedicated to bringing AI safely and effectively into healthcare.

TECHNICAL SKILLS

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|--------------|-----------|-----------|---------|
| • Docker | • FastMCP | • PyTorch | • GCP |
| • Kubernetes | • Next.js | • WandB | • AWS |
| • MLflow | • Rust | • SLURM | • Neo4J |

EXPERIENCE

KRV LABS — Co-Founder and Chief Science Officer

September 2026 - Present

- Leading **clinical trial patient-discovery**, including longitudinal search across diagnoses, treatments, and notes. Our platform is powered by topological and geometric *representation learning* and *EHR foundation models*.

INDEPENDENT — Machine Learning and AI Consultant (Part time)

June 2023 - August 2026

- Led **healthcare ML projects**, including a UTI prediction model for CHOC Children's—deployed via *FastAPI* backend and *Node.js* mobile interface used by urologists in-clinic.
- Developed **full-stack HIPAA-compliant platform** for Compass Health's healthcare concierge, *streamlining integrated workflows for doctors, patients, and care coordinators* to enhance healthcare delivery.

HELMHOLTZ MUNICH — Doctoral Researcher

August 2022 - August 2026

- Leading research projects in **geometric deep learning**, developing open-source frameworks (Rings, Scott) for graph structure evaluation and representation robustness. *Publications in NeurIPS 2023 and ICML 2025*.
- Leading **representation learning** publications (Presto) using topology and geometry to analyze embedding sensitivity in large language models. *Published at ICML 2024*.
- Collaborating on high-impact, international projects including **coal-plant phaseout strategies** and the efficacy of **care delivery networks** (Thema, Apparent). *Publications in Nature Energy & Communication in CS*.

EDUCATION

TECHNICAL UNIVERSITY OF MUNICH — Dr. rer. nat. in Mathematics

August 2022 - September 2026

- Research in mathematics for machine learning with three first-author publications (NeurIPS, ICML) and two applied papers in high impact journals (energy & medicine). *Magna Cum Laude*.

CHAPMAN UNIVERSITY — Msc. Computational Data Sciences and Machine Learning

August 2020 - June 2022

- Coursework in *AI, statistics, and advanced computing* (GPA 3.8). Conducted research in topology and category theory; served as **grant-funded research scientist** at CHOC's Computational & Data Science Research Team.

UNIVERSITY OF CALIFORNIA, BERKELEY — B.Sc. in Mathematics (Honors) and Astrophysics

August 2015 - December 2019

- Rigorous coursework in physics & mathematics. Completion of **honors thesis** in category theory & logic. Contributions in *theoretical astrophysics* (w/ R. Anantua) and *observational cosmology* (w/ A. Filippenko).

PUBLICATIONS

NeurIPS 2023 (Curvature Filtrations for Graph Generative Model Evaluation); ICML 2024 (Mapping the Multiverse of Latent Representations); ICML 2025 (Principled Evaluations of Graph-Learning Datasets); Nature Communications in Computer and Information Science (Physician Referral Networks via Ricci Curvature); Nature Energy (Strategies to Accelerate US Coal Power Phaseout)